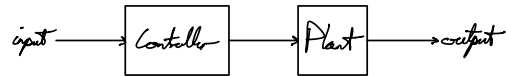
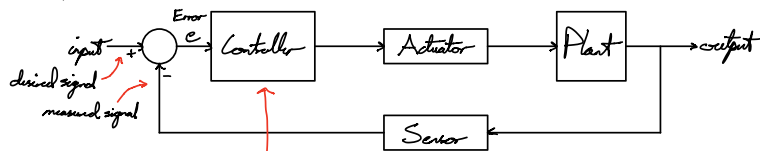


- a Controlled Object is called *system, plant or process*
- Automatic Control allows for *convenience, safety, precision and increased efficiency*
- Open-Loop Control: output is controlled without sensors $\rightarrow$ no feedback



- Closed-Loop Control: compare output with signal input to minimise error signal



- Controller Design Process
 1. Modelling
 2. Analysis
 3. Design

Modelling

- Laplace Transform
 - Unit Step Function: $F(s) = \int_0^{\infty} 1 \cdot e^{-st} dt = \frac{1}{s}$
 - Unit Ramp Function: $F(s) = \int_0^{\infty} t \cdot e^{-st} dt = \frac{1}{s^2}$
 - Unit Impulse Function: $F(s) = \int_0^{\infty} \delta(t) e^{-st} dt = 1$

- Table of Transforms

$$\mathcal{L} \left\{ \frac{t^{n-1}}{(n-1)!} \right\} = \frac{1}{s^n}$$

$$\mathcal{L} \left\{ \frac{e^{-at} t^{n-1}}{(n-1)!} \right\} = \frac{1}{(s+a)^n}$$

$$\mathcal{L} \{ \cos \omega t \} = \frac{s}{s^2 + \omega^2}$$

$$\mathcal{L} \{ \sin \omega t \} = \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L} \{ e^{at} f(t) \} = \bar{f}(s-a)$$

$$\mathcal{L} \{ H(t-a) f(t-a) \} = e^{-sa} \bar{f}(s)$$

$$\mathcal{L} \{ \delta(t-a) f(t) \} = f(a) e^{-sa}$$

$$\mathcal{L} \{ t^k f(t) \} = (-1)^k \frac{d}{ds^k} \bar{f}(s)$$

- Final Value Theorem

- poles of $sF(s)$ are in *open LHP*

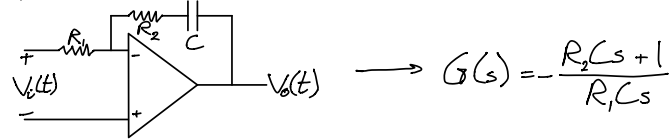
$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

- Initial Value Theorem

- poles of $sF(s)$ can be located *anywhere*

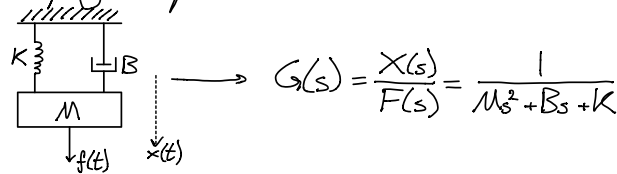
- Models for Physical Systems

- Electrical: Operational Amplifier

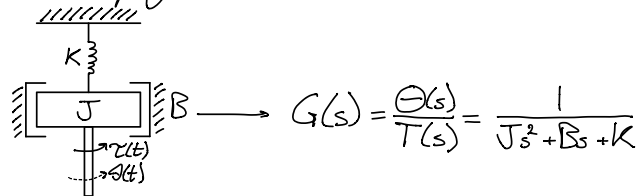


- Mechanical

- Linear: Mass/Spring/Damper



- Rotational: Inertia/Spring/Friction



- State Space Representation

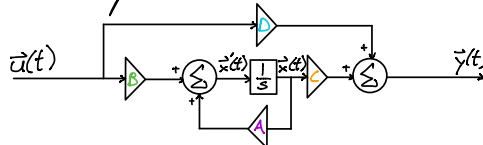
- model of physical system as a set of **input**, **output**, and **state** variables related by **first-order** differential equations

- knowledge of input $[u_1(t), u_2(t)]$ and initial state values $[x_1(t_0), \dots, x_n(t_0)]$ suffices to determine future output $[y_1(t), y_2(t)]$ and state variables



- State: set of first order differential equations written in terms of state variables

- CT System State-Space LTI Model



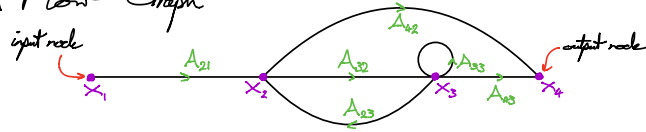
$$\dot{\bar{x}}(t) = \mathbf{A}\bar{x}(t) + \mathbf{B}\bar{u}(t) \rightarrow \text{State Differential Equation}$$

$$\bar{y}(t) = \mathbf{C}\bar{x}(t) + \mathbf{D}\bar{u}(t) \rightarrow \text{Response Equation}$$

• Output Response: $\vec{y}(t) = \mathcal{C}e^{At}\vec{x}_0 + \mathcal{C} \int_0^t e^{A(t-\tau)} \mathcal{B}u(\tau) d\tau + \mathcal{D}u(t)$

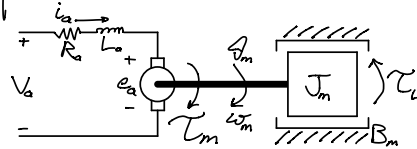
$$\mathcal{L}\{\vec{y}(t)\} = \mathcal{C}(sI - A)^{-1}\vec{x}_0 + \underbrace{[\mathcal{C}(sI - A)^{-1}\mathcal{B} + \mathcal{D}]}_{\text{Transfer Function: } \mathcal{T}(s)} U(s)$$

• Signal Flow Graph



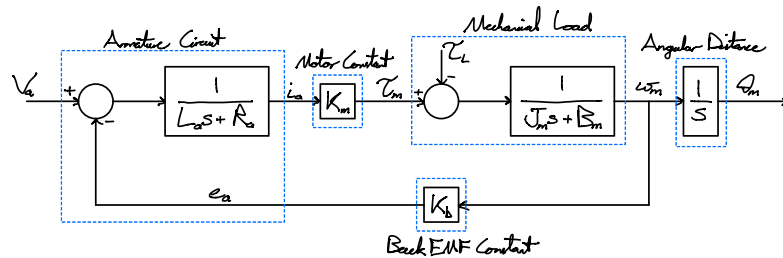
- Nodes connected by Branches with Gains
- Path Gain is product of branches between nodes
- Loop Gain is product of gains of a loop

• Model of DC Motor



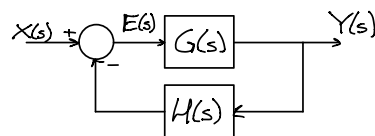
$$V_a = R_a i_a + L_a \frac{di_a}{dt} + e_a$$

$$J_m \omega_m' = T_m - B_m \omega_m - T_L$$



• Feedback Reduction

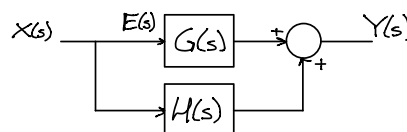
• Negative Feedback



$$Y(s) = \frac{\overbrace{G(s)}^{\text{forward gain}}}{\underbrace{1 + G(s)H(s)}_{\text{open-loop TF / loop gain}}} \cdot X(s)$$

denominator = 0
Characteristic Equation

• Forward Loop



$$Y(s) = (G(s) + H(s)) \cdot X(s)$$

• Linearization

- a system is linear if it is both *Homogeneous* & *Additive*

$$S(ax) = aS(x) \quad S(x+y) = S(x) + S(y)$$

- to linearize a system at $x = x_0$: $M\ddot{x}(t) = F(t) - Kx(t) - x^3(t)$

1. identify input and output variables: $x(t), y(t)$

2. express in form: $f(x, \dot{x}, \ddot{x}, \dots, y, \dot{y}, \ddot{y}, \dots) = 0$

3. find *operating point*: $(x, y) = (x_0, y_0)$

4. find *first order* Taylor Series Expansion at *operating point*:

$$\begin{aligned} f(x, \dot{x}, \ddot{x}, \dots, y, \dot{y}, \ddot{y}, \dots) &\approx f(x_0, 0, 0, \dots, y_0, 0, 0, \dots) \\ &+ \left. \frac{\partial f}{\partial x} \right|_q (x - x_0) + \left. \frac{\partial f}{\partial \dot{x}} \right|_q (\dot{x} - \dot{x}_0) + \left. \frac{\partial f}{\partial \ddot{x}} \right|_q (\ddot{x} - \ddot{x}_0) + \dots \\ &+ \left. \frac{\partial f}{\partial y} \right|_q (y - y_0) + \left. \frac{\partial f}{\partial \dot{y}} \right|_q (\dot{y} - \dot{y}_0) + \left. \frac{\partial f}{\partial \ddot{y}} \right|_q (\ddot{y} - \ddot{y}_0) + \dots \end{aligned}$$

5. change variables to *perturbation variables*:

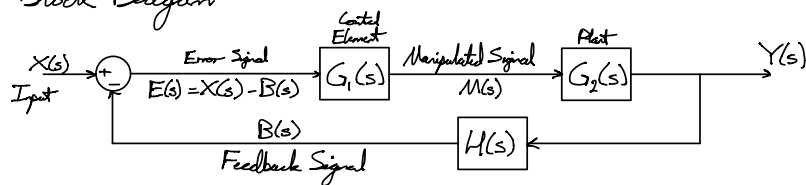
$$\tilde{x} = (x - x_0), \quad \tilde{\dot{x}} = (\dot{x} - \dot{x}_0), \quad \tilde{\ddot{x}} = (\ddot{x} - \ddot{x}_0), \dots$$

$$\tilde{y} = (y - y_0), \quad \tilde{\dot{y}} = (\dot{y} - \dot{y}_0), \quad \tilde{\ddot{y}} = (\ddot{y} - \ddot{y}_0), \dots$$

6. rewrite functions in standard linear ODE form:

$$\left. \frac{\partial f}{\partial \tilde{x}} \right|_q \tilde{x} + \left. \frac{\partial f}{\partial \tilde{\dot{x}}} \right|_q \tilde{\dot{x}} + \left. \frac{\partial f}{\partial \tilde{\ddot{x}}} \right|_q \tilde{\ddot{x}} + \dots = - \left. \frac{\partial f}{\partial \tilde{y}} \right|_q \tilde{y} - \left. \frac{\partial f}{\partial \tilde{\dot{y}}} \right|_q \tilde{\dot{y}} - \left. \frac{\partial f}{\partial \tilde{\ddot{y}}} \right|_q \tilde{\ddot{y}} - \dots$$

• Block Diagram



$$\frac{Y(s)}{X(s)} = \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s)} \rightarrow \text{closed loop transfer function}$$

$$\frac{Y(s)}{E(s)} \rightarrow \text{feed-forward transfer function}$$

Analysis

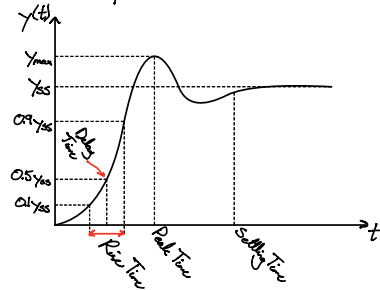
Stability

- Bounded Input Bounded Output stability
 - any **bounded input** generates a bounded output
- Asymptotic stability
 - any initial conditions generates output that converges to zero
- Both conditions satisfied if poles of transfer function are in **open LHP**
- Marginal stability
 - if transfer function has:
 - no pole in open RHP
 - at least one pole on $j\omega$ -axis
- Routh-Hurwitz Criterion
 - determines whether all poles of linear system have negative real parts
 - determines the number of poles in open RHP
 - Routh Array of: $\mathcal{Q}(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$

s^n	a_n	a_{n-2}	a_{n-4}	a_{n-6}	...	
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	a_{n-7}	...	
s^{n-2}	b_1	b_2	b_3	b_4	...	$b_i = \frac{a_{n-2i} \cdot a_{n-1} - a_n \cdot a_{n-(2i+1)}}{a_{n-1}}$
s^{n-3}	c_1	c_2	c_3	c_4	...	$c_i = \frac{a_{n-(2i+1)} \cdot b_1 - a_{n-1} \cdot b_{i+1}}{b_1}$
s^{n-4}	d_1	d_2	d_3	d_4	...	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$d_i = \frac{b_{i+1} \cdot c_1 - b_1 \cdot c_{i+1}}{c_1}$
s^2	k_1	k_2				
s^1	l_1					
s^0	m_1					

- number of roots in open RHP equals number of sign changes in **first column**
- all roots are in LHP if all constants (a_x) have same signs (+/-)
- Row of Zeros: if there exists a row (s^x) containing only zeros; the polynomial has roots on $j\omega$ -axis or RHP
 - replace row of zeros with derivative of **auxiliary polynomial**
 - if sign change(s) at RoZ $\rightarrow$ RHP pole(s), otherwise $\rightarrow$ $j\omega$ -axis pole(s)
- Ex: if s^{n-3} is row of zeros; $P(s) = b_1 s^{n-2} + b_2 s^{n-4} + b_3 s^{n-6} \dots$
- Epsilon Method: if there exists a 0 in the first column; replace it with a small positive number (O^+)

• Time Response: $y(t) = y_t(t) + y_{ss}(t)$

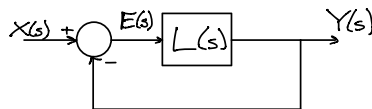


Steady State Error: $e_{ss} = 1 - y_{ss}$

Percent Overshoot: $\frac{y_{max} - y_{ss}}{y_{ss}}$

• 2% Settling time: time to settle within 2% of y_{ss}

• Steady State Error of Feedback System



$$e(t) = x(t) - y(t)$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{sX(s)}{1 + L(s)}$$

• Error Constants: represents system's ability to reduce e_{ss}

• Step-error (distance-error) constant: $K_p = \lim_{s \rightarrow 0} L(s) \rightarrow e_{ss} = \frac{1}{1 + K_p}$

• Ramp-error (speed-error) constant: $K_v = \lim_{s \rightarrow 0} sL(s) \rightarrow e_{ss} = 1/K_v$

• Parabolic-error (acceleration-error) constant: $K_a = \lim_{s \rightarrow 0} s^2 L(s) \rightarrow e_{ss} = 1/K_a$

• system becomes more accurate when $e_{ss} \rightarrow 0$; error constant $\rightarrow \infty$

- for $K_p \rightarrow \infty$; $L(s)$ must have at least 1 integrator
- for $K_v \rightarrow \infty$; $L(s)$ must have at least 2 integrators
- for $K_a \rightarrow \infty$; $L(s)$ must have at least 3 integrators

• 1st Order System: $G(s) = \frac{K}{\tau s + 1}$ ← DC Gain

- 2% settle time: 4τ
- 5% settle time: 3τ

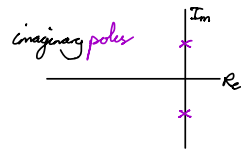
← Time Constant

• 2nd Order System: $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ ← Undamped Natural Frequency

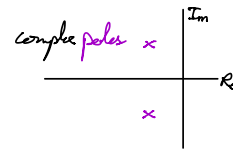
- 2% settle time: $4/\zeta\omega_n$
- 5% settle time: $3/\zeta\omega_n$

← Damping Ratio

• Undamped: $\zeta = 0$



• Underdamped: $0 < \zeta < 1$



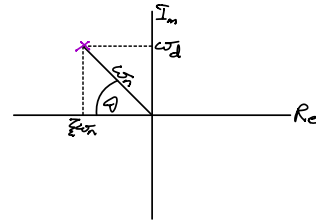
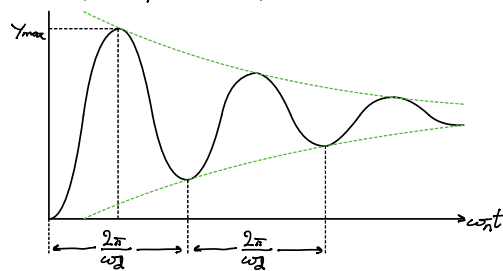
• Critically Damped: $\zeta = 1$



• Overdamped: $\zeta > 1$



• Step Response Properties



Peak Value: $y_{max} = 1 + e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$

Damping Frequency: $\omega_d = \omega_n \sqrt{1-\zeta^2}$

Envelope: $1 \pm \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}$

Time Constant: $\tau = \frac{1}{\zeta\omega_n}$

Peak Time: $\tau_p = \frac{\pi}{\omega_d}$

Percent Overshoot: $e^{-\pi/\tan\phi}$

• Nyquist
a polar plot of the frequency response of a linear system

• Sketching Points

1. start of plot: $\omega = 0$
2. end of plot: $\omega = \infty$
3. real axis intercept: $\text{Im}[G(j\omega)] = 0$
4. imaginary axis intercept: $\text{Re}[G(j\omega)] = 0$

• Stability Criterion

stable if: $Z = P + N = 0$

Z: # of CL poles in open RHP

P: # of OL poles in open RHP

N : # of *clockwise* encirclement of -1 - # of *counterclockwise* encirclement of -1

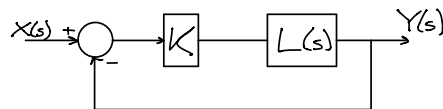
- Relative Stability

- Gain Margin: $GM = 20 \log_{10} \frac{1}{|L(j\omega_p)|}$; $\angle L(j\omega_p) = -180^\circ$
- Phase Margin: $PM = \angle L(j\omega_g) + 180^\circ$; $|L(j\omega_g)| = 1$

Design

- Root Locus

- graphical representation of how pole locations of closed-loop system change as proportional gain (K) varies



- Sketching

- mark open-loop poles ($\times$) and zeros ($\circ$)
- real-axis points* exist left of odd numbered root and right of even numbered root
 - root locus *originates* at poles and *terminates* at zeros/infinity-zeros
 - # infinity zeros = # poles - # zeros

3. draw **asymptotes** originating at centroid (α)

- # asymptotes = $\text{deg}(\text{denominator}) - \text{deg}(\text{numerator}) = r$
- angles of asymptotes = $\frac{\pi}{r}(2k+1)$, $\{k | 0 \leq k \leq (r-1), k \in \mathbb{Z}\}$
- intersection of asymptotes = $\frac{\sum p_k - \sum z_k}{r} = \alpha$

4. **breakaway** points originate where two pole branches meet
break-in points terminate where two zero branches meet

- amongst roots of $L'(s) = 0$; if $K = -\frac{1}{L(s_0)} > 0$, then s_0 is break point
- **imaginary-axis intersection** point: use Routh Array to determine maximum K value

5. determine **departure angle** of a complex pole (p_x) using angle condition:

$$\angle(s_0 - p_x) = 180^\circ + \sum_{i=1}^m \angle(s_0 - z_i) - \sum_{i=1, i \neq x}^n \angle(s_0 - p_i)$$

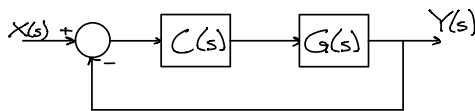
$\hookrightarrow$ approximately equal to departure angle of p_x for $s_0 \doteq p_x$

determine **arrival angle** of a complex zero (z_x) using angle condition:

$$\angle(s_0 - z_x) = 180^\circ - \sum_{i=1, i \neq x}^m \angle(s_0 - z_i) + \sum_{i=1}^n \angle(s_0 - p_i)$$

$\hookrightarrow$ approximately equal to arrival angle of z_x for $s_0 \doteq z_x$

- Lead/Lag Compensator: $C_{\text{lead}}(s) \cdot C_{\text{lag}}(s)$
- improves undesirable frequency response of system



- desired pole: $s_d = -\zeta \omega_n + j \omega_n \sqrt{1 - \zeta^2}$

- **lead** compensator moves centroid **left**
 - improves transient response
 - improves stability

$$C_{\text{lead}}(s) = K \frac{s + z_{\text{lead}}}{s + p_{\text{lead}}}$$

$$1. z_{\text{lead}} = |\text{Re}(s_d)|$$

$$2. p_{\text{lead}} \leftarrow \angle \left(\frac{s_d + z_{\text{lead}}}{s_d + p_{\text{lead}}} \right) = 180^\circ - \angle G(s_d)$$

$$3. K = |L(s_d)|^{-1}$$

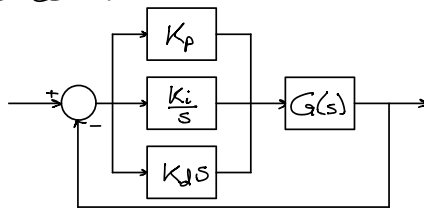
- *lag* compensator moves centroid *right*
reduce steady state error

$$C_{lag}(s) = \frac{s_d + z_{lag}}{s_d + p_{lag}} \doteq 1$$

1. given maximum e_{ss} and p_{lag}

$$2. K_v = \lim_{s \rightarrow 0} sG(s)C_{lead}(s)C_{lag}(s) = \infty \cdot \frac{z_{lag}}{p_{lag}}$$

- PID Controller



$$C(s) = K_p + \frac{K_i}{s} + K_d s = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

- Integral Control: higher K_i decreases rise time and increases oscillations
- Derivative Control: higher K_d increases damped response

- Ziegler-Nichols Tuning

- Step Response Method

	K_p	T_i	T_d
P	$1/a$		
PI	$0.9/a$	3τ	
PID	$1.2/a$	2τ	$\tau/2$

- Ultimate Sensitivity Method

	K_p	T_i	T_d
P	$0.5K_c$		
PI	$0.4K_c$	$0.8T_c$	
PID	$0.6K_c$	$0.5T_c$	$T_c/8$